RELATIVE CYCLES ARE STABLE UNDER THE STANDARD OPERATIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Relative cycles (relative-cycles chapter, Definition definition-relative-cycles, canonical tag 0H50). cited at statement lines 966-967. Defines the relative cycle condition the lemma operates on.
- Specialization maps commute with flat pullback (relative-cycles chapter, Lemma lemma-specialization-flat-pullback, canonical tag 0H4L). cited at proof line 973. Covers flat pullback, hence restriction as its special case.
- Specialization maps commute with proper pushforward (relative-cycles chapter, Lemma lemma-specialization-proper-pushforward, canonical tag 0H4M). cited at proof line 976. Covers proper pushforward.

Source reference. Original definition 0H50: sources/stacks-chow/tags.txt, source lines 21370–21370. Cited at statement lines 966-967: defines the relative cycle condition.

Source reference. Original lemma 0H4L: sources/stacks-chow/tags.txt, source lines 21356–21356. Cited at proof line 973: covers flat pullback and restriction.

Source reference. Original lemma 0H4M: sources/stacks-chow/tags.txt, source lines 21357–21357. Cited at proof line 976: covers proper pushforward.

Lemma 0.2. *Let α be a relative r-cycle on X/S as in Definition 0H50. Then any restriction, base change, flat pullback, or proper pushforward of α is a relative r-cycle.*

Proof. For flat pullback use Lemma 0H4L. Restriction is a special case of flat pullback. To see it holds for base change use that base change is transitive. For proper pushforward use Lemma 0H4M. □

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